Benchmarking Quantum computers

Quantum volume and cousins

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Benchmarking Quantum computers

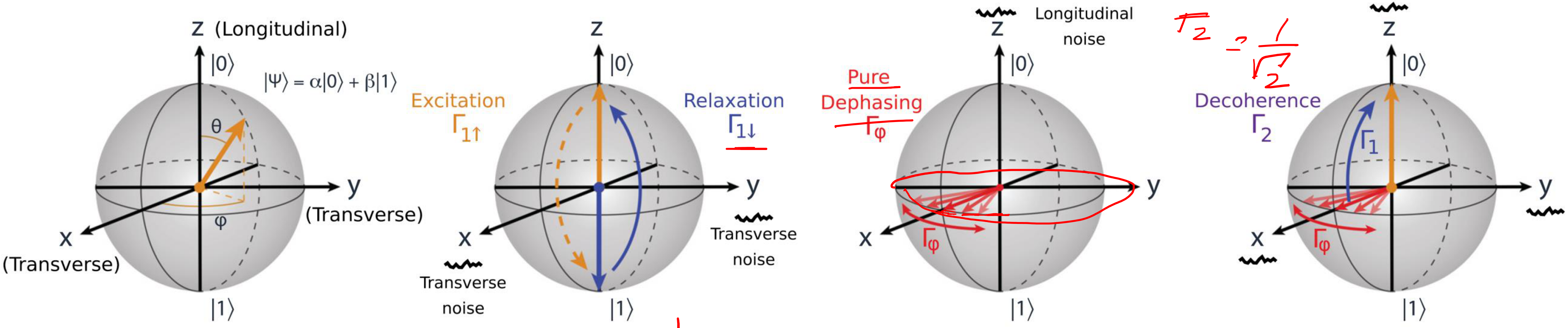
Level 3: Full circuit level, agonistic (Quantum Volume)

Level 2: Gate level, circuit subsystem (Random Clifford Gates)

Level 1: Qubit device level (T1, T2, Cross talk, etc)

Benchmarking Qubits

Level 1: Qubit device level (T1, T2, Cross talk, etc)



$|1\rangle$ T_1

$|0\rangle$

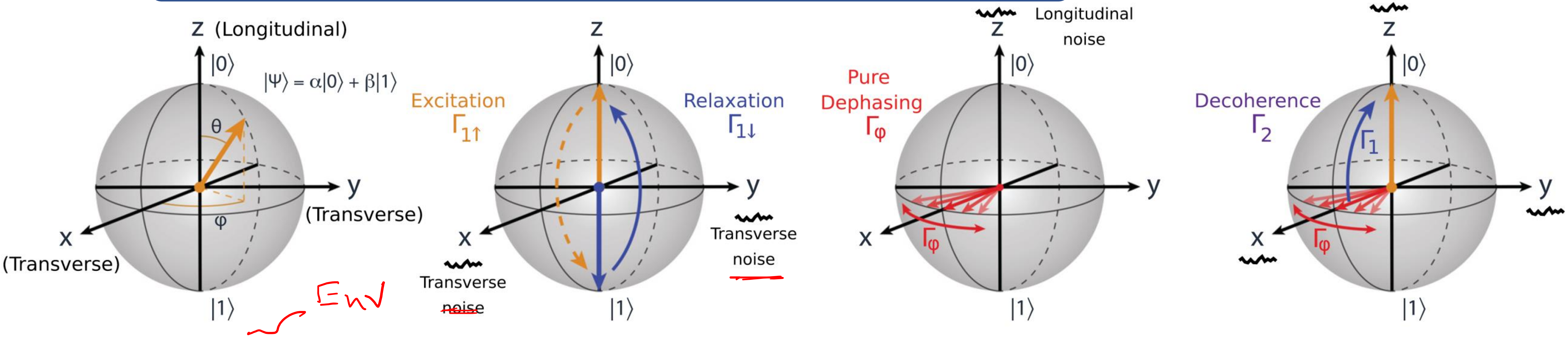
$|1\rangle$ $T_1 = \frac{1}{\Gamma_{12}}$

$|0\rangle$

Ref: P. Krantz *et. al*, Appl. Phys. Rev. 6, 021318 (2019)

Benchmarking Qubits

Level 1: Qubit device level (T1, T2, Cross talk, etc)



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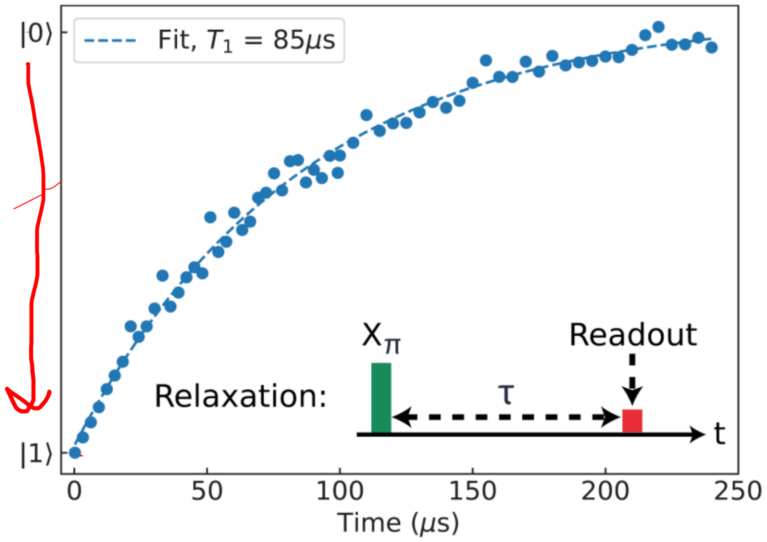

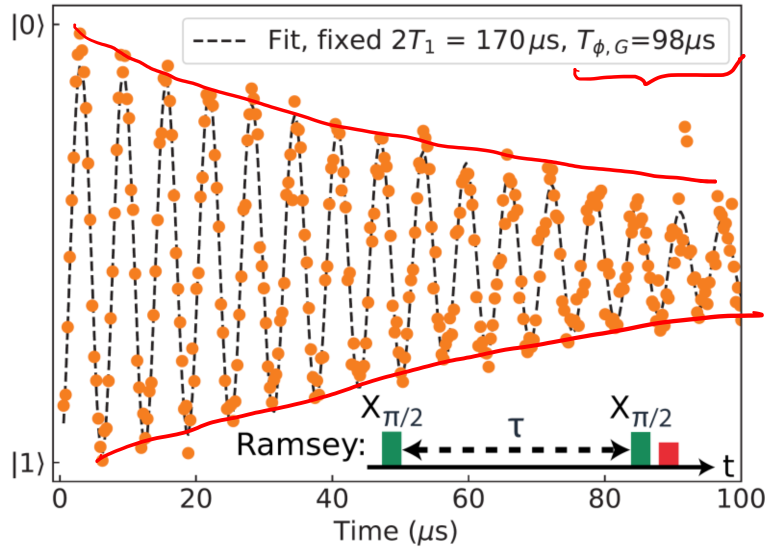

T1, T2

Ref: P. Krantz *et. al*, Appl. Phys. Rev. 6, 021318 (2019)

Benchmarking Qubits

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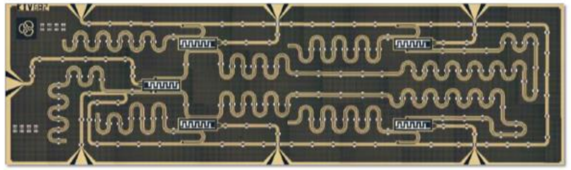
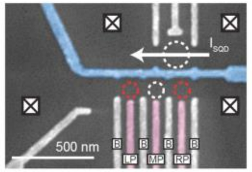
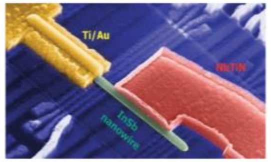
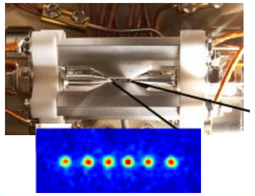
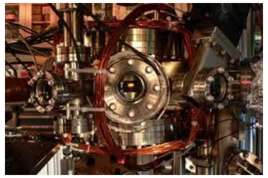
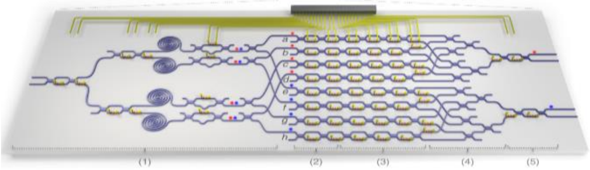



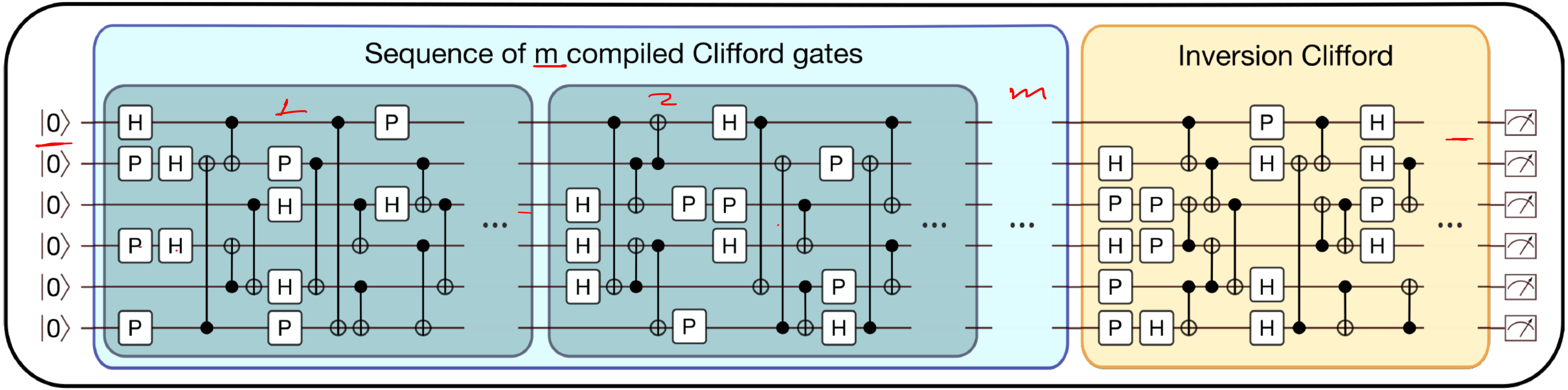


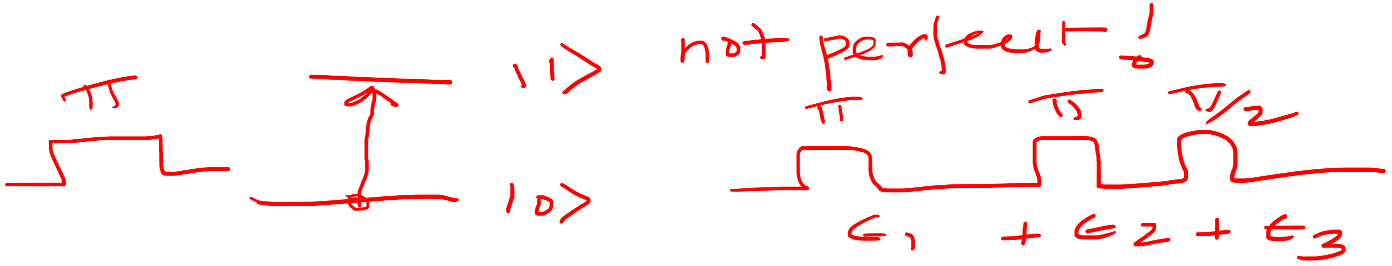




	Superconducting	Spin	Topological	Ion Trap	Neutral Atom	Photonic
	IBM Rigetti Google Alibaba	Intel	Microsoft	IonQ Honeywell AQT	ColdQuanta Atom Computing	Psi Quantum Xanadu ORCA Computing
T_1	?	?		?		
T_2	?	?		?		

Benchmarking circuit subsystem

Level 2: Gate level, circuit subsystem (Random Clifford Gates)





Ref: Timothy J. Proctor, arXiv:1807.07975v3 (2012)

Benchmarking circuit subsystem

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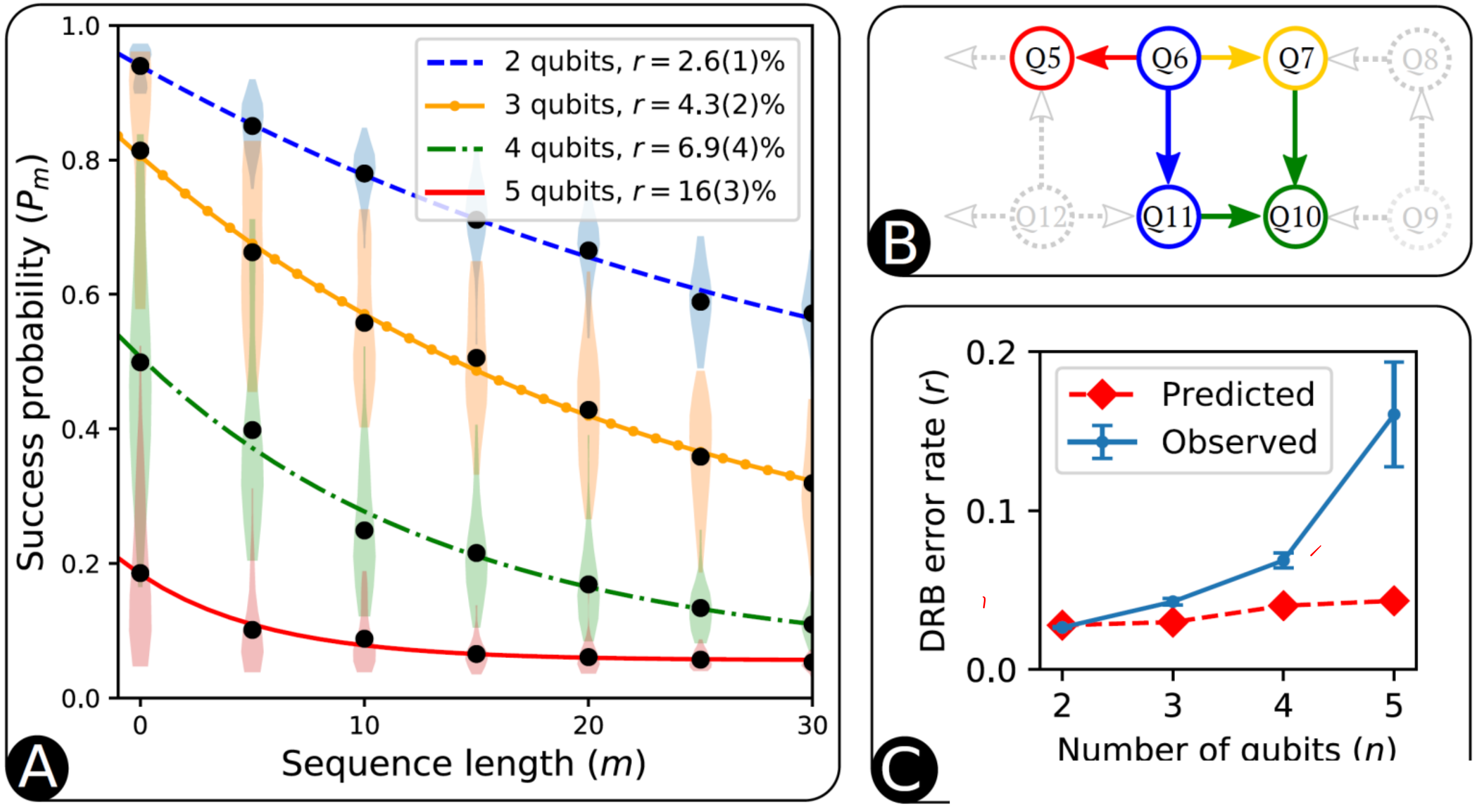

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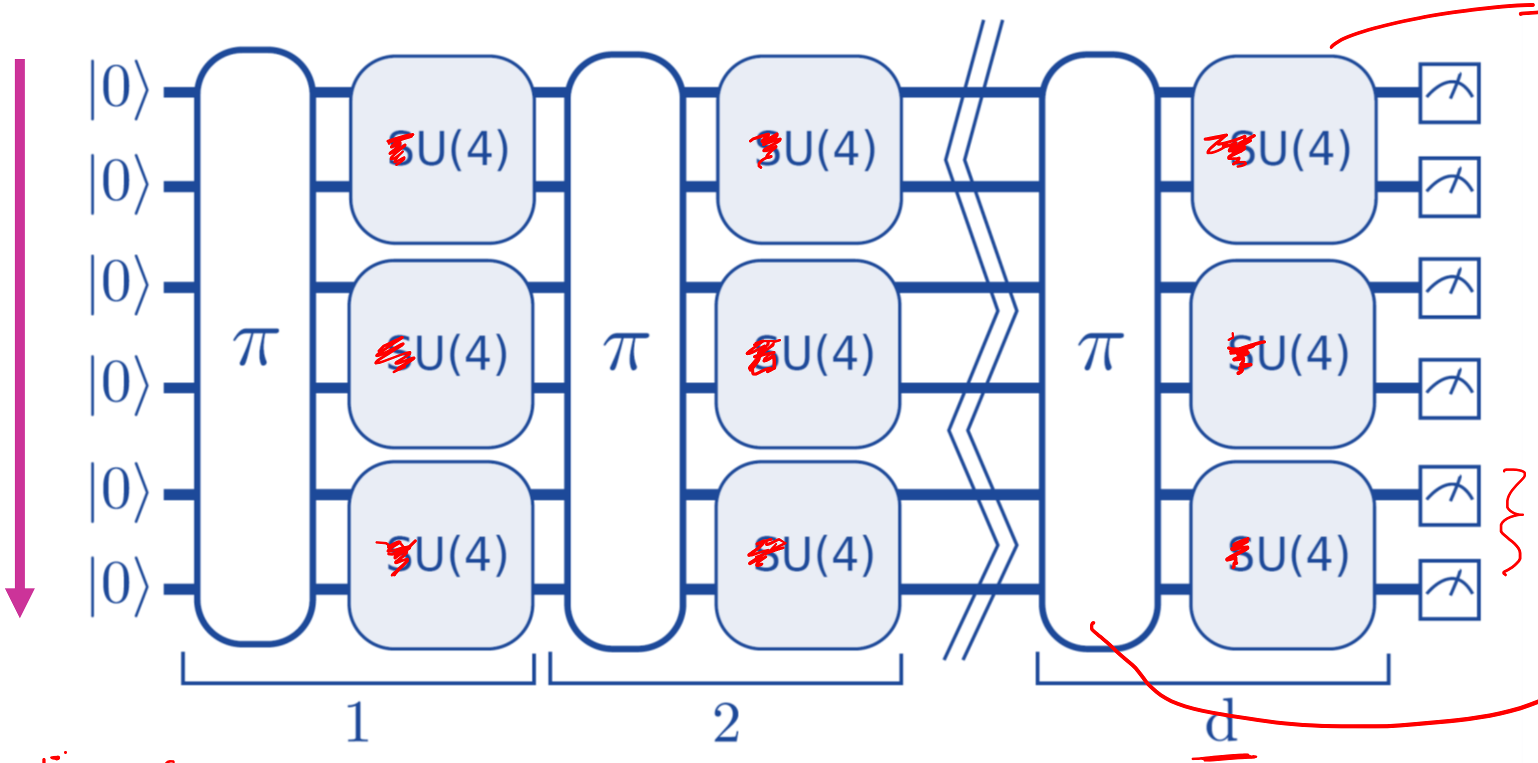
Full circuit benchmarking

Level 3: Full circuit level, agonistic (Quantum Volume)

n Qubits



Random 2 qubit gates

Uniformly Random Permutations between qubits

Space time

$$V = n.d \sim 1/\epsilon_{eff}$$

$$\boxed{V_Q = 2^{\min(n,d)}}$$

Ref: Andrew W. Cross *et.al.* ,Phys. Rev. A 100 , 032328 (2019)

Full circuit benchmarking

Level 3: Full circuit level, agonistic (Quantum Volume)

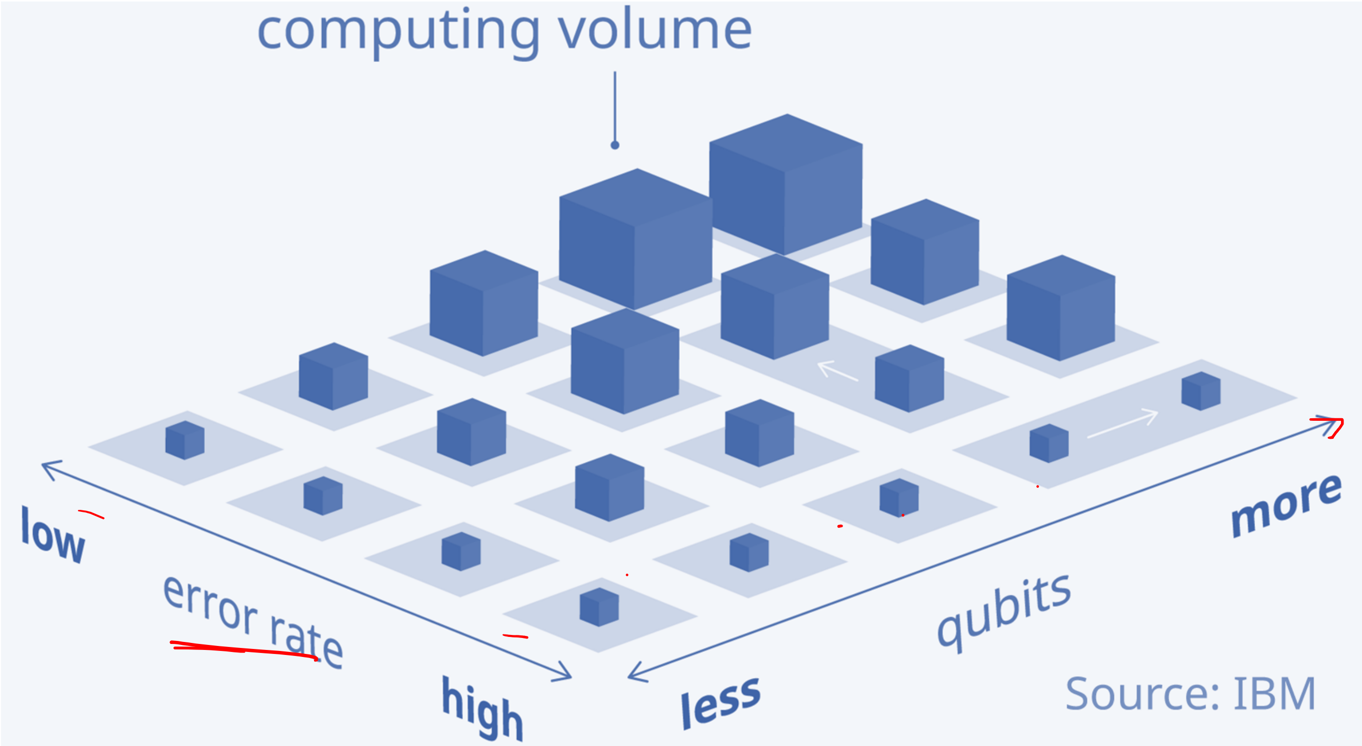
Qubits: More is better

Connectivity: More is better

Gate Set: More is better

Errors : Less is better

Decoherence: Less is better



V_Q & ion Q
IBM, Google

$$V = n.d \sim 1/\epsilon_{eff} \quad \left[V_Q = 2^{\min(n,d)} \right]$$

Ref: Andrew W. Cross *et.al.* ,Phys. Rev. A 100 , 032328 (2019)

Thank you for your attention!